BM
4/5/22

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

PRECIPITATION SAMPLES (TUES)

DATE COLLECTED: 29 MAR 2022 Initials TW

Conversion Factors
10 in. Bulk Rain Funnel: NET divided by 51.1 = catch
N/A if Overfilled
12 in. Bulk Snow Bucket: NET divided by 69.3 = catch

SITE	TIME (EST)	COLLECTOR VOLUME (ml)	COLLECTOR CATCH (mm)	ETI CATCH (mm)	Coll EFF. (% of ETI)
RG-22	1135	1110	16.0		
RG-1	1020	1291	18.6		
RG-4	1000	1295	18.7		
RG-23	0825	1732	25.0		

Volumes Collected

		RG-22	RG-1	RG-4	RG-23
Gross	g	2070	2251	2257	2693
Tare	g	960	960	962	961
Net	g	1110	1291	1295	1732

SITE CONTAMINATION
(0 = none 1 = low ----------------------------> 5 = high)

SITE	Fine PM	Coarse PM	Bird FECES	Insects	Color	GB Ring	dBASE FIELD code?	SAMPLE REJECTED @LEVEL =	Database CODES checked ?
RG-22	4 Sand	2	0	0	0	2			
RG-1	2+	0	0	0	0	1			
RG-4	3	1	0	0	0	1			
RG-23	3	2	0	0	0	2			

Precipitation amount
1.02 inches / 25.9 mm
ETI collector

ADDITIONAL COMMENTS: Precip was mixed rain / snow, which melted / refroze when collected. Forest Service snow scale data - 0-42 cm D, 0-24 cm SWE. SG-22 = 0", SG-1 = 0", SG-4 = 3", SG-23 = 15"

Applicable dBASE FIELD codes: (See SOP Manual)
970 = Obvious Contamination: Sample REJECTED
969 = Possible Contamination Check Return Analyses
955 = QNS: Quantity Not Sufficient for analyses (< 100 mL NET catch)

HUBBARD BROOK ECOSYSTEM STUDY **SAMPLE NOTES** WINDY AND COLD! 19 °F.

Streamwater Samples Date Collected: 29 MAR 2022 Initials. TW/GW

SITE	TIME (EST)	TEMP (°C)	GAGE HT.	HYDROGR L/R/F/PK	COLOR (0 = none:1 = low---->5 = high)	SEDIMENT	REMARKS	dBASE FIELD code?
WS-1	1020	–	FZN	L	1-	0		
WS-2	1015	–		L	0	0		
WS-3	0945	–		L	1-	0		
WS-4	1040	–		L	0	0		
WS-5	0920	–		L	0	1		
WS-6	0930	–		L	1-	0		
WS-7	0845	0.1		L	0	1		
WS-8	0835	0.1		L	1	0		
WS-9	0815	0.1	FZN	L	2	0		
Hubbard BK	1115	–		L	1	1		
Mlake Outlet	1205	2.0°		F	1-	0	OTB, Ice-free staff.	

ADDITIONAL COMMENTS: Winter returns, with re-icing events. Hoods removed, so all V-notches full of ice - No GH data. All streams (and weirs) mostly or fully ice-covered. Dusting of fresh snow SF, NF - still 100% SC. Thermistor failed on SF side. Red maples bud burst @ STA. Drove the Ranger, tracked vehicle, No more snowmobiling on SF side. GW harvested 2 bags of litter from weirs 1, 3, 9.

Applicable dBASE FIELD codes: 911= High Flow Event in Progress, 912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)

955 = QNS: Quantity Not Sufficient for analyses

HUBBARD BROOK ECOSYSTEM STUDY

METROHM / 3 STAR LAB ANALYSES

DATE RUN: 30 MAR 2022 RUN By: TW

Meter CALIBRATION: See Calibration form for Standard Procedure

Conductivity Correction Factor = 1.12 x at: 18.7 °C Lab Temp

Precipitation

Samples:	DATE COLLECTED	Metrohm pH	3 Star pH	SpCond @ 25°C	Remarks Codes
STATION	29 MAR 2022	5.43	5.45	5.3	
RG-1	↓	5.44	5.47	5.0	
RG-4	↓	5.48	5.48	4.9	
RG-23	29 MAR 2022	5.38	5.39	4.5	

Streamwater

Samples	DATE COLLECTED	Metrohm pH	Metrohm ANC	ANC end pt. pH	3 Star pH	SpCond @ 25°C	Remarks Codes
WS-1	29 MAR 2022	5.49	+5.2	5.16	5.50	13.3	
WS-2	↓	5.56	+1.7	5.26	5.56	8.0	
WS-3	↓	5.69	+11.2	5.24	5.68	9.2	
WS-4	↓	5.95	+15.5	5.12	5.94	9.2	
WS-5	↓	5.46	+2.9	5.09	5.46	9.5	
WS-6	↓	5.41	+4.9	5.11	5.45	10.1	
WS-7	↓	5.96	+16.7	5.16	5.96	9.5	
WS-8	↓	5.81	+14.3	5.11	5.79	9.8	
WS-9	↓	4.77	−25.3	4.46	4.79	13.4	
Hubbard BK	↓	6.19	+22.8	5.20	6.18	10.5	
Mlake Outlet	29 MAR 2022	6.35	+68.1	5.07	6.31	30.0	

ADDITIONAL COMMENTS:

Applicable codes: 911= High Flow Event in Progress;

955= QNS· Quanity Not Sufficient for analyses(<100 mL NET catch)

912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)

DIC-[date] xls <-save as [date]= mm-dd-yy

HBEF Stream and M.Lake Routine Sample DIC Data

date: **30MAR2022** Samples collected 03/29/2022

Aqueous STANDARDS

type	DIC set =	peak ht mm	adjusted mm	ATT -X
DIW blank	0	10	0	16
	0	11	0	16
	0	11	0	16
100 µM	100	30	20	16
	100	30	19	16
	100	30	19	16
200 µM	200	50	40	16
	200	51	40	16
	200	50	39	16
400 µM	400	97	87	16
	400	92	81	16
	400	95	84	16

Statistics *Values from matrix at M5 -->*

slope =	4 732
slope ±	0 078
Y-int =	5 8
Y-int ±	3.7
R sq =	0 9973

Laboratory Conditions

Lab Temp =	18.7	°C
Barom Pressure =	770	mm Hg
Analyst's Initials.	TW	45%RH
NaCO3 Standards·		
0 1M Stock Dated .	3/10/22	
0 01M Working Solution Dated·	3/10/22	

Samples	inj 1 1	inj 2 2	inj 3 3	ATT -X	Avg Peak	±SD Peak	Samples	DIC µM/L	Calc err±	NOTES 120 mA current
ML-70	39	40	37	16	38 7	1.5	ML-70	**183**	7	
	0	0	0	16	0.0	0.0	0	**0**	0	
	0	0	0	16	0.0	0.0	0	**0**	0	
	0	0	0	16	0.0	0.0	0	**0**	0	
	0	0	0	16	0.0	0.0	0	**0**	0	
W-1	18	19	20	16	19.0	1.0	W-1	**90**	5	
W-2	14	14	16	16	14 7	1 2	W-2	**69**	5	
W-3	13	13	14	16	13 3	0 6	W-3	**63**	3	
W-4	14	14	15	16	14.3	0 6	W-4	**68**	3	
W-5	11	12	12	16	11.7	0.6	W-5	**55**	3	
W-6	11	11	12	16	11.3	0 6	W-6	**54**	3	
W-7	13	13	14	16	13 3	0 6	W-7	**63**	3	
W-8	13	13	14	16	13 3	0 6	W-8	**63**	3	
W-9	10	10	11	16	10 3	0.6	W-9	**49**	3	
HBk	14	14	14	16	14.0	0.0	HBk	**66**	0	
	0	0	0	16	0.0	8.0	0			
							0			
Lab Air	29	30	30	16	29.7	0.6	Lab Air	**140**	3	
Outside Air	12	13	13	16	12.7	0.6	Outside Air	**60**	3	
Gas 1000	25	25	25	16	25.0	0.0	Gas 1000	**118**	0	std=83.3
Gas 3000	70	71	73	16	71 3	1.5	Gas 3000	**338**	7	*std=246 3*